ANDREAS OIKONOMOU

Electrical and Electronics Engineer · PhD Candidate · TEE-TCG Licensed Engineer

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PROFILE

Electrical and Electronics engineer, PhD candidate working at the intersection of quantum physics, space optics, and hardware engineering. Currently a member of the CMW group (BEC Lab & Space Optics Lab) at IESL-FORTH, developing precision optical systems for space applications in collaboration with ESA and DLR. Licensed electrical engineer (TEE-TCG) with practical experience across embedded systems, PCB design, SDR, and lab-grade instrumentation. Available for select B2B hardware consulting and licensed engineering sign-offs in Greece.

EXPERIENCE

PhD Candidate
University of Crete — Dept. of Materials Science & Engineering *2025 – Present · Heraklion, Greece*

- Doctoral research focusing on advanced space optics and quantum technologies, building on ongoing work at IESL-FORTH.

Electronics Engineer — CMW Group (BEC Lab & Space Optics Lab)
IESL-FORTH · ESA & DLR Collaborator *2024 – Present · Heraklion, Greece*

- Member of the CMW group, working across both the BEC Lab and Space Optics Lab at IESL-FORTH.
- Developing a fibered Raman Beam Combiner for space quantum sensors (ESA/RamanBC): achieved >90% coupling efficiency on Zerodur optical breadboards; designed TVAC thermal control loops with NI-9149/9219 hardware and custom LabVIEW data-logging for orbital qualification testing.
- Designed and fabricated precision modulation coils for the BEC3 machine — custom 3D Fusion 360 formworks, flat copper wire winding, strict dimensional tolerances for TOP trap magnetic field homogeneity.
- Built a 10-channel bi-stable optical shutter driver for laser experiments; characterised mechanical response (~25ms) and verified stable frequency response from 1 Hz to 15 kHz.
- Producing technical documentation and reports to ESA standards; presenting results at international conferences.

R&D Electronics Engineer — Internship
TEM Company *2023 – 2024 · Denmark*

- Embedded systems development using STM32 microcontrollers in a professional R&D setting.
- PCB design with KiCAD/OrCAD Allegro; prototype assembly, testing, and iteration in a multinational team.

Research Member
Electronic Devices & Materials Research Lab, UniWA *2020 – 2023 · Athens, Greece*

- Built a portable autonomous EM spectrum analyser (RTL-SDR + Raspberry Pi 4 + GPS) with custom Python FFT library for wideband RF scanning and spatial logging. Validated against known FM broadcast frequencies.

Licensed Electrical Engineer — Freelance
Technical Chamber of Greece (TEE-TCG) *2025 – Present · Greece*

- TEE-certified engineer authorised to issue legally binding signatures, conduct on-site inspections, and certify electrical projects under HD 384 / ELOT 60364 standards.
- Available for B2B hardware consulting: PCB design, embedded firmware, lab automation, and ESA-standard technical documentation.

EDUCATION

PhD Candidate, Materials Science & Engineering
University of Crete — *2025 – Present · Heraklion, Greece*

MEng, Electrical & Electronics Engineering
University of West Attica — *2020 – 2025 · Athens, Greece*

- *Thesis: Integrated system for wideband EM radiation recording — SDR-based approach (RTL-SDR, RPi 4, GPS, Python FFT library)*

CONFERENCE POSTERS

- **Development of Stable and Precise Optical Breadboards: Raman Beam Combiner**
  A. Oikonomou, M. Georgousi, P. Examilioti, W. von Klitzing
  7th ESA Quantum Technology Conference, FORTH-ITE, Heraklion
- **Quantum-Limited Measurements and Control of Atom Number in Ultracold Ensembles**
  P. Rivero-Antúnez, A. Oikonomou, K. Makris, R. Thampy, V. Pathak, M. Chairetis, W. von Klitzing
  7th ESA Quantum Technology Conference / FORTH-QuTech, Heraklion
- **Quantum Memory Prototype for Space**
  M. Georgousi, P. Rivero-Antunez, A. Oikonomou, R. Thampy, K. Makris, W. von Klitzing
  FORTH-QuTech Center for Quantum Science & Technologies, Heraklion
- **Development and Optimization of a Laser System for Bose-Einstein Condensate Experiments**
  K. Makris, A. Oikonomou, V. K. Pathak, V. P. Veetil, R. P. Thampy, P. Rivero-Antunez, W. von Klitzing
  FORTH-QuTech Center for Quantum Science & Technologies, Heraklion

TECHNICAL SKILLS

Python / MATLAB / C	STM32 / Arduino / RPi	PCB Design (KiCAD/OrCAD)
GNU Radio / SDR / GQRX	Laser & Optics Systems	Zerodur / Space Optics
LabVIEW / NI Hardware	VISA / Lab Instrumentation	TVAC Testing
3D Printing / Fusion 360	Spectrum & Signal Analysis	Linux Admin (RHEL)

LANGUAGES

English C2 Proficiency — University of Michigan
Russian Native
Greek Native

SUMMER SCHOOLS & CERTIFICATION

TEE-TCG Licensed Engineer
Technical Chamber of Greece — Electrical Engineering

ELI Summer School 2023
ELI Beamlines Facility, Dolní Břežany, Czech Republic